

Development of car fuel types in Switzerland



Juan Ruiz, June 2026

- How do fuel-type shares vary across Swiss cantons?
- How does the canton of Aargau compare to others in battery-vehicle adoption in the most current year?
- How have fuel types developed over time?
(out of scope for today)

Data

Scope

- Data from 2015 to 2025
- Date per Canton
- Only Passenger cars
- 10 different fuel types

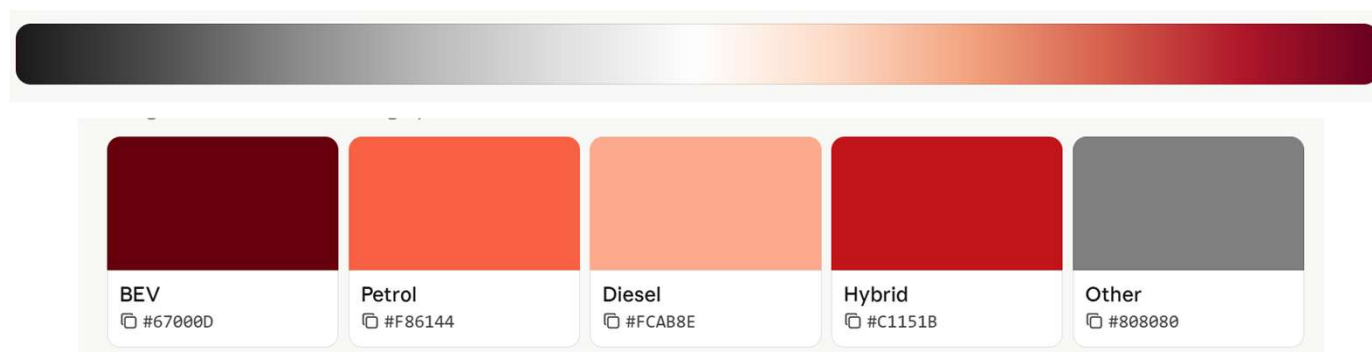
Source



Canton	Type of holder	Vehicle group / type	Fuel	TIME_PERIOD	OBS_VALUE
Obwalden	Natural person	Passenger car	Petrol: hybrid electric vehicle (HEV)	2025	1519
Fribourg	Natural person	Passenger car	Plug-in hybrid electric vehicle (PHEV): diesel	2025	75
St. Gallen	Legal entity	Passenger car	Diesel: conventional	2017	17619
Valais	Natural person	Passenger car	Diesel: conventional	2024	53246
Vaud	Natural person	Passenger car	Petrol: hybrid electric vehicle (HEV)	2024	30638

General approach

- **Munzner's Why / What / How**
- **Category design** *Merged the 5 hybrid variants into one to reduce cognitive load*
- **No norm for fuel colors** *EN 16942 uses symbols, not color for accessibility*
- **Evluation 5+ participants.** *Following Nielsen's of 5 users*

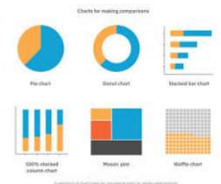


How do fuel-type shares vary across Swiss cantons?

Ask Google AI for a map solution

I want to plot the proportion of vehicles of each of the five fuel types in each canton on a map. What chart should I use?

To plot the proportion of five different vehicle fuel types in each canton, you should use a **bivariate map that embeds a Pie Chart or Stacked Bar Chart in each canton.** esri Esri +1



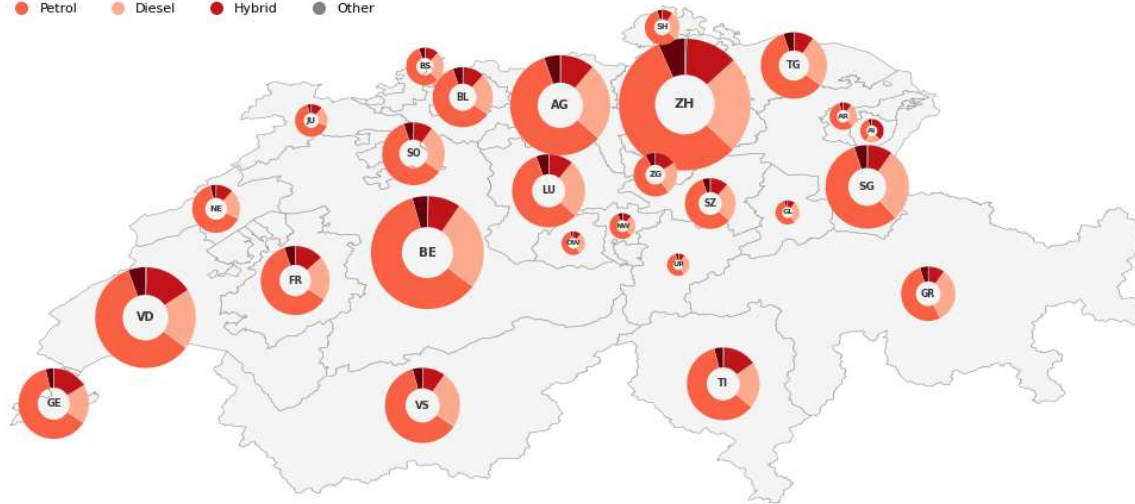
Because displaying five categories simultaneously across 26 Swiss cantons can get visually crowded, here are the best charting techniques ranked by readability:

1. Pie Chart Map (Best for Whole-to-Part)

Fleet composition and size by canton 2025

Donut size scales with total registered cars while colour shows the fuel type share.

● BEV ● Petrol ● Diesel ● Hybrid ● Other



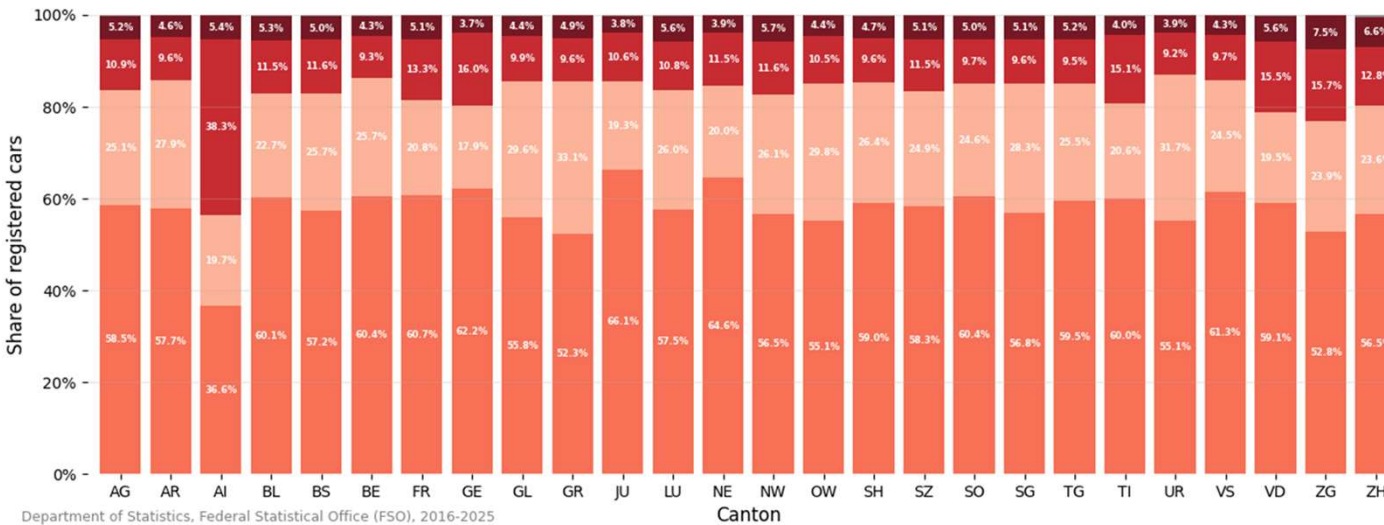
Department of Statistics, Federal Statistical Office (FSO), 2016–2025

- **Angles are hard to compare**
(Cleveland & McGill 1984).
- **Viewers must hold values in memory to compare two cantons. Higher cognitive load**
(Franconeri, Padilla, & Shah 2021)
- **If Bars placed on a map, unaligned position and therefore not as accurate than aligned** (Munzner 2014).
- **Evulation:** Geography ≠ the task. The question is "how do shares *compare*?"

Fuel type share of registered cars per canton 2025

Percentage labels inside each segment; segments smaller than 6% are unlabeled.

Petrol Diesel Hybrid BEV Other



- Only the bottom segment shares a common axis (Schwabish 2021; Munzner 2014).
- 5 categories sits at the upper boundary of the 3–5 segment recommendation (Munzner 2014).
- Labels become cluttered quickly ~20 bars is at the recommended threshold (Schwabish 2021).
- Evaluation: number of elements are overwhelming.

How does the canton of Aargau compare to others in battery-vehicle adoption?

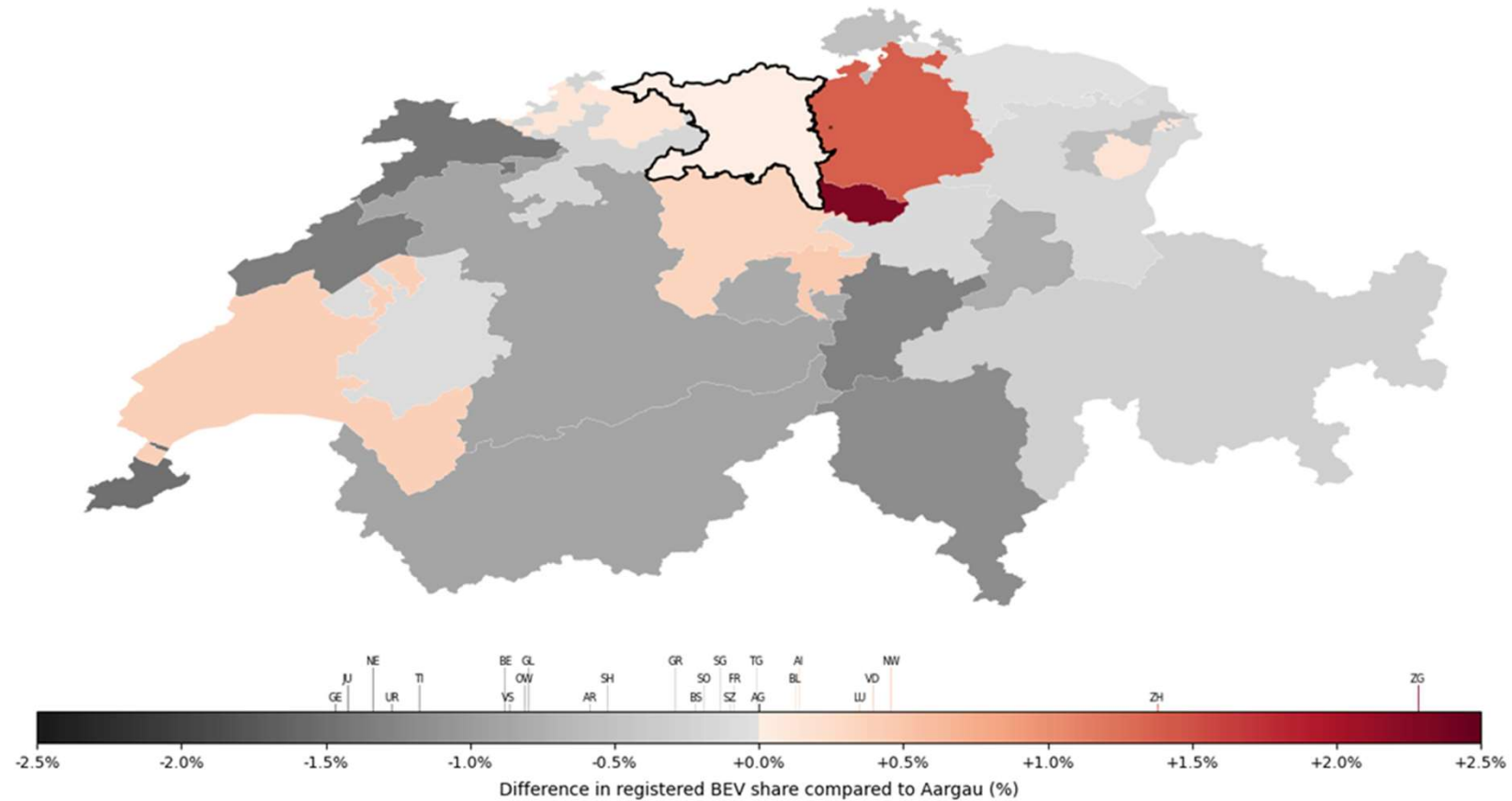
Share of newly registered Battery Electric Vehicles (BEV) per canton compared to Aargau share of 5.21%, which serves as the zero reference.



- Add the ranking into the color bar

Share of registered Battery Electric Vehicle in each canton compared to Aargau 2025

Share of newly registered Battery Electric Vehicles (BEV) per canton compared to Aargaus share of 5.21%, which serves as the zero reference.

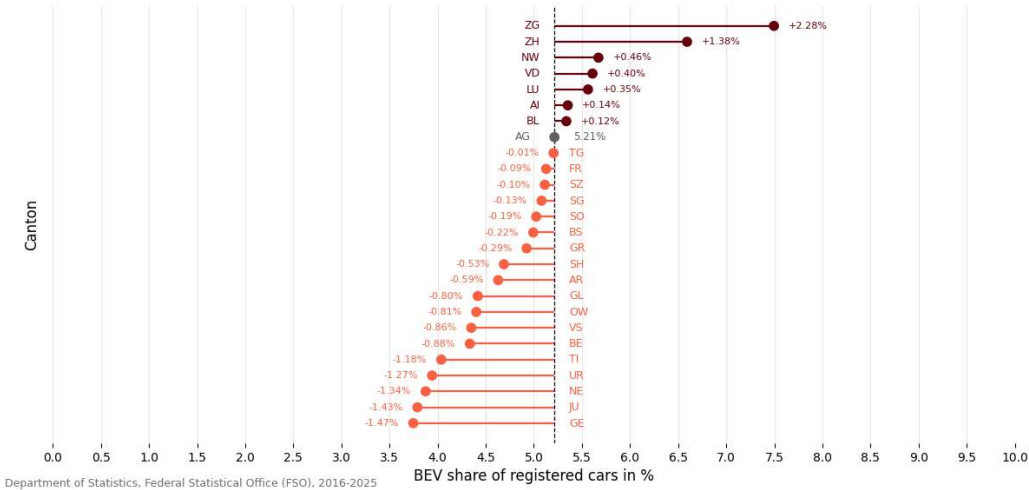


Department of Statistics, Federal Statistical Office (FSO), 2016–2025. Canton boundaries: Swisstopo.

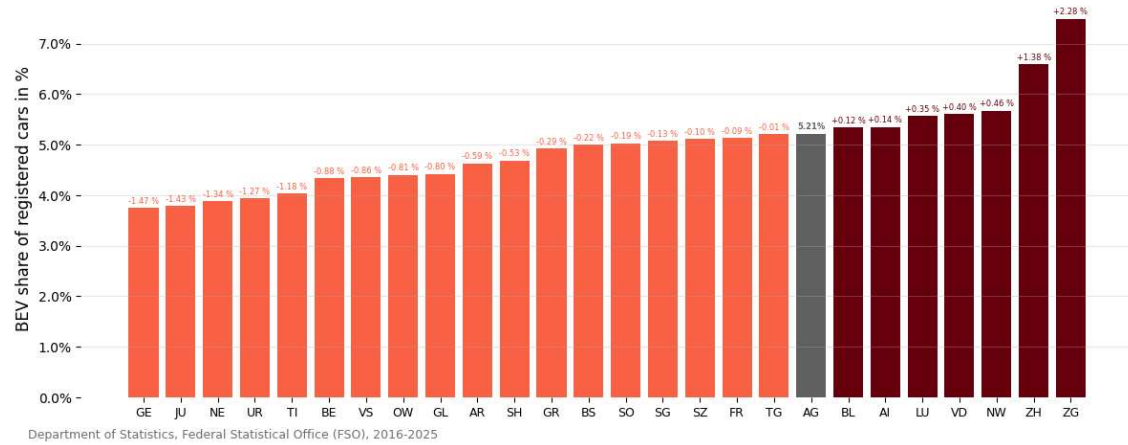
Share of registered Battery Electric Vehicle in each canton compared to Aargau 2025

Share of newly registered Battery Electric Vehicles (BEV) per canton compared to Aargaus share of 5.21%.

a

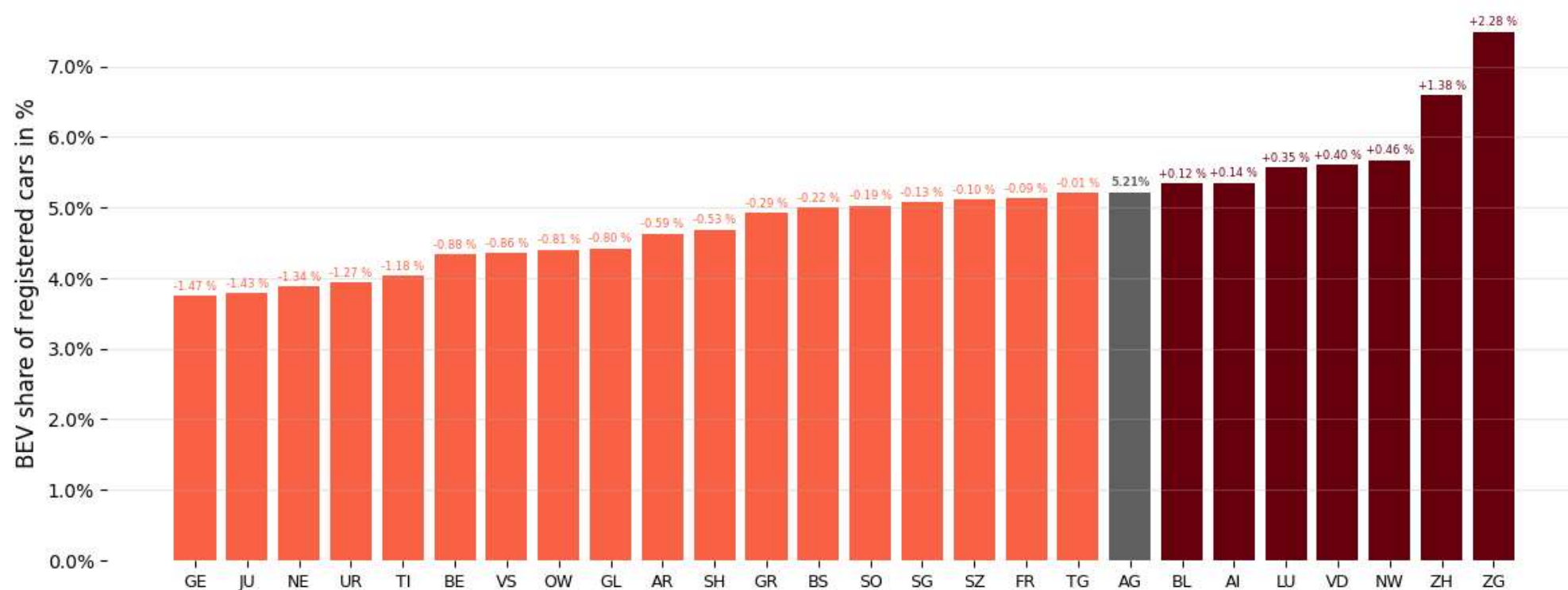


b



Share of registered Battery Electric Vehicle in each canton compared to Aargau 2025

Share of newly registered Battery Electric Vehicles (BEV) per canton compared to Aargaus share of 5.21%.



Department of Statistics, Federal Statistical Office (FSO), 2016-2025

Takeaways

About the Data

- Basel Land has a higher BEV share than Basel Stadt
- AI has the highest hybrid share.

For Data Viz

- Ensure the encoding matches the task.
- Don't trust everything that an AI or Google suggests.

References

- **Cleveland, W. S., & McGill, R. (1984).** *Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods*. Journal of the American Statistical Association, 79(387), 531-554. <https://doi.org/10.1080/01621459.1984.10478080>
- **Franconeri, S. L., Padilla, L. M., & Shah, P. (2021).** *The Science of Visual Data Communication: What Works*. Proceedings of the National Academy of Sciences, 118(8). <https://doi.org/10.1073/pnas.2005254118>
- **Munzner, T. (2014).** *Visualization Analysis and Design*. CRC Press. <https://doi.org/10.1201/b17511>
- **Nielsen, J. (2000).** *Why You Only Need to Test with 5 Users*. Nielsen Norman Group. <https://www.nngroup.com/articles/why-you-only-need-to-t>
- **Schiewe, J. (2019).** *Empirical Studies on the Visual Perception of Spatial Patterns in Choropleth Maps*. KN - Journal of Cartography and Geographic Information, 69, 217-228. <https://doi.org/10.1007/s42489-019-00026-y>
- **Schwabish, J. (2021).** *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*. Columbia University Press.
- **Swiss Federal Statistical Office. (2026).** *Bestand der Strassenfahrzeuge nach Fahrzeuggruppe und Fahrzeugart*. Retrieved 2026-03-05 from <https://stats.swiss>